

Physics-Informed Knowledge Distillation for Real-Time State-of-Charge Estimation of LiFePO₄ Batteries on Automotive Microcontrollers

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Abstract—Precise state-of-charge (SOC) estimation in Lithium Iron Phosphate (LiFePO₄ or LFP) batteries remains an enduring industrial challenge for automotive battery management systems (BMS) due to the ultra-flat open-circuit voltage (OCV) plateau ($dV_{ocv}/dSOC \approx 0.1\text{--}0.5$ mV/%) and protracted non-linear solid-phase relaxation dynamics. While heavy data-driven ensembles and electrochemical partial differential equation (PDE) solvers capture these path-dependent transitions, their computational complexity and excessive memory footprints prohibit deterministic real-time execution on resource-constrained automotive microcontrollers. In this study, we propose MicroPhys-BMS, an end-to-end Physics-Informed Machine Learning Knowledge Distillation (PIML-KD) framework that compresses complex non-linear electro-thermal relaxation dynamics into an ultra-compact student Micro-MLP architecture ($12 \rightarrow 32 \rightarrow 16 \rightarrow 1$, 961 parameters) for ISO 26262 ASIL-D compliant microcontrollers. The framework extracts a 12-dimensional domain-guided physical feature vector encoding Arrhenius reaction kinetics, dynamic ohmic overpotential, logarithmic solid-phase diffusion scaling, and differential relaxation gradients. A 300-tree gradient-boosted decision tree (LightGBM) teacher supervises the student network via a composite physics-calibrated objective combining adaptive Huber distillation, task-level mean squared error, and quadratic physical boundary constraints enforcing $SOC \in [0.0, 1.0]$. Validated against extensive Stanford LFP operational telemetry, the proposed student Micro-MLP achieves a nominal 25°C case-split MAE of 0.454% and RMSE of 1.236% ($R^2 = 0.9986$), outperforming the LightGBM teacher (MAE = 0.488%). Under unseen cell generalization tests, the model records SOC MAEs of 2.48% on cell YX07 and 3.35% on cell YX08, reducing estimation errors by 42.3% and 22.1% relative to the Stanford baseline benchmark target (4.30%). For short-term rest horizons, the framework enables robust Coulomb counting re-initialization at 30 s (MAE = 1.170%) and converges to 0.339% at 300 s. Furthermore, the model demonstrates high resilience against analog front-end (AFE) voltage noise (MAE = 1.336% under ± 5.0 mV perturbation) and multi-temperature shifts (MAE = 2.689% at 35°C), with 95.94% of cumulative error distribution remaining within $\leq 2.0\%$. Synthesized into a MISRA-C:2012 compliant static C header without dynamic heap allocation (malloc), the deployed model requires only 3.75 kB Flash, 192 Bytes SRAM, and an execution latency of ≈ 12 μ s on an ARM Cortex-M4 at 160 MHz, achieving over 99.7% memory compression compared to standard tree ensembles.

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Index Terms—Automotive microcontrollers, battery management system (BMS), knowledge distillation (KD), lithium iron phosphate (LFP) battery, physics-informed machine learning (PIML), state-of-charge (SOC).

Nomenclature

Variables and Physical Descriptors	
$t, V_{\text{mea}}, V_{10}$	Rest time (s), measured voltage, V at $t = 10$ s (V).
$SOC_{\text{mea}}, y_{\text{std}}$	Coulomb-counted SOC, predicted student SOC.
I_m, R, I_{flag}	Pre-rest current (A), dynamic resistance (Ω), direction (± 1).
$T_{\text{env}}, \Delta T$	Ambient temperature and thermal gradient ($^{\circ}\text{C}$).
ΔV_{10}	Relaxation overpotential gradient, $V_{\text{mea}}(t) - V_{10}$.
$\tau_{\text{log}}, P_{\text{ohm}}$	Diffusion scale $\ln(t+1)$; ohmic dissipation $R \cdot I_m$.
Φ_{Arr}	Arrhenius factor $\exp(-E_a/(R_g T_K))$; $E_a = 35.0$ kJ/mol.
$X_{\text{raw}}, X_{\text{PIML}}$	Raw 9-D and physics-informed 12-D feature vectors.
$y_{\text{true}}, y_{\text{teach}}$	Ground-truth SOC and teacher soft targets.
$\mathcal{L}_{\text{PIML}}$	Composite physics-informed distillation loss.
$\alpha, \beta, \lambda_{\text{mse}}, \lambda_{\text{loss}}$	Loss weights (0.40, 0.015, 0.50, 0.02).
$\mathbf{W}_l, \mathbf{b}_l, \sigma_{\text{SiLU}}$	Layer l weights/biases; SiLU $x/(1 + e^{-x})$.
$T_{\text{exec}}, M_{\text{Flash}}, M_{\text{SRAM}}$	Latency (μ s), Flash and SRAM footprints.

Acronyms and Abbreviations

AFE,	Analog Front-End; Automotive Safety Integrity
ASIL-D	Level D (ISO 26262).
BMS,	Battery Management System; Vehicle Control
VCU	Unit.
ECM,	Equivalent Circuit Model; Extended Kalman
EKF,	Filter; Gradient Boosted Decision Tree.
GBDT	
LFP,	Lithium Iron Phosphate; Open-Circuit Voltage;
OCV,	Physics-Informed Machine Learning.
PIML	
MAE,	Mean Absolute Error; Root-Mean-Square Error;
RMSE,	Worst-Case Execution Time.
WCET	
MISRA-C	Motor Industry Software Reliability Association (2012).

I. Introduction

LITHIUM-ION batteries, particularly Lithium Iron Phosphate (LiFePO₄ or LFP), have established decisive dominance as the preferred electrochemical energy storage technology for commercial electric vehicles (EVs), urban transit buses, and heavy-duty grid energy storage systems [1]–[4].

A. Industrial Context and the LFP Flat-Plateau Challenge

Driven by their exceptional inherent thermal runaway resilience, operational cycle life exceeding 3000 full equivalent cycles, non-toxic cobalt- and nickel-free chemistry, and superior raw material cost-effectiveness, LFP battery

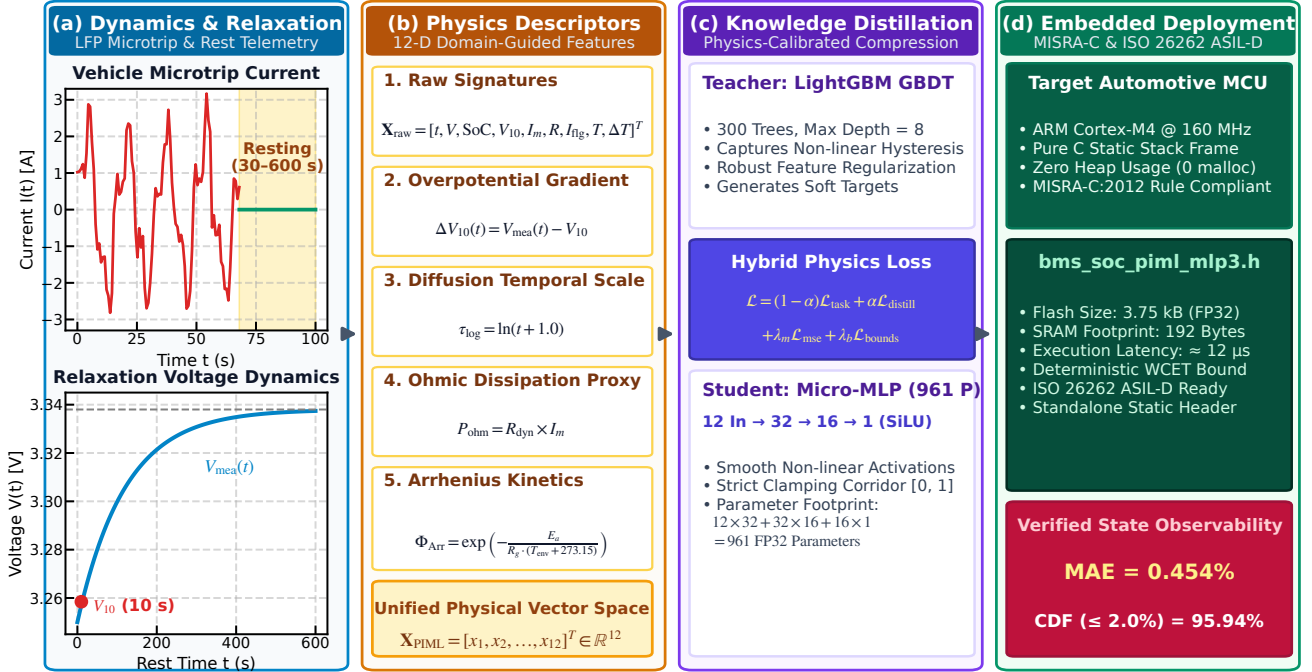


Fig. 1: Architectural overview of the proposed MicroPhys-BMS framework. (a) Real-world operational vehicle microtrip current profile and transient relaxation voltage dynamics (30–600 s). (b) Domain-guided physical feature engineering extracting 12-dimensional electro-thermal descriptors ($\mathbf{X}_{PIML}$). (c) Hybrid knowledge distillation compressing a 300-tree LightGBM ensemble into a 961-parameter student Micro-MLP under multi-objective physics-calibrated loss. (d) Automotive embedded deployment pipeline yielding a MISRA-C compliant, zero-malloc static C header (bms_soc_piml_mlp3.h) for deterministic execution on automotive microcontrollers.

packs have achieved dominant market penetration in commercial transportation electrification [1]–[3], [5], [6]. Despite these compelling safety and economic advantages, deploying LFP battery chemistries in advanced automotive battery management systems (BMS) introduces severe, non-linear state observation challenges that are fundamentally distinct from nickel-manganese-cobalt (NMC) or lithium titanate (LTO) chemistries [7]–[10].

The central impediment to reliable on-board state-of-charge (SOC) tracking in LFP cells resides in the thermodynamic flat open-circuit voltage (OCV) plateau, which spans an extensive operational corridor from approximately 20% to 80% SOC [9], [11]. From a solid-state electrochemical perspective, lithium deintercalation and insertion in the olivine LiFePO_4 lattice proceed via a first-order phase transformation between two distinct ordered crystal phases: the fully lithiated triphylite (LiFePO_4) phase and the fully delithiated heterosite (FePO_4) phase [9]. In the thermodynamic miscibility gap, the chemical potential of lithium ions μ_{Li^+} remains strictly constant across the phase boundary. According to the Nernst-Planck thermodynamic relation:

$$V_{ocv}(\text{SOC}, T_K) = -\frac{\mu_{\text{Li}^+}^{\text{cathode}}(\text{SOC}, T_K) - \mu_{\text{Li}^+}^{\text{anode}}(\text{SOC}, T_K)}{F}, \quad (1)$$

where $F = 96485.33$ C/mol denotes Faraday's constant and T_K is absolute temperature. Consequently, as illus-

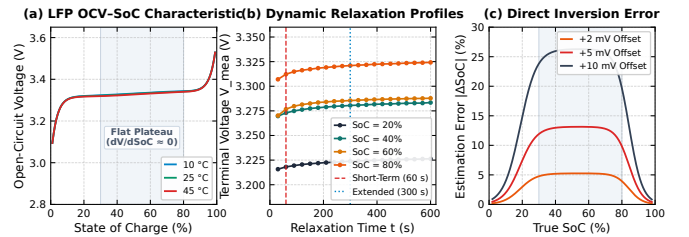


Fig. 2: Electrochemical characteristics and observability dilemma of LFP cells. (a) Static V_{ocv} -SOC characteristic across 10°C, 25°C, and 45°C highlighting the flat plateau. (b) Dynamic terminal voltage relaxation trajectories across varied SOC levels (20%, 40%, 60%, and 80%). (c) Direct thermodynamic inversion error under typical analog front-end (AFE) sensing offsets (+2 mV, +5 mV, and +10 mV).

trated in Fig. 2(a), the incremental derivative of the equilibrium open-circuit voltage with respect to SOC is virtually zero throughout this wide domain:

$$\left. \frac{\partial V_{ocv}(\text{SOC}, T_K)}{\partial \text{SOC}} \right|_{\text{SOC} \in [0.20, 0.80]} \approx 0.10\text{--}0.50 \text{ mV}/\%. \quad (2)$$

Compounding this flat derivative is significant path-dependent electrochemical hysteresis between charging and discharging branches, which arises from mechanical coherency strain energy and phase-boundary interfacial resistance [7], [9]. Because the static voltage curve is nearly

horizontal, any direct numerical inversion of the measured terminal voltage V_{mea} to deduce battery state exhibits extreme mathematical ill-conditioning. As demonstrated in Fig. 2(c), an analog front-end (AFE) measurement offset of merely +5.0 mV induces catastrophic SOC estimation errors exceeding 20%–30% across the plateau domain.

Furthermore, dynamic vehicle operation introduces prolonged, non-linear voltage relaxation tails following current pulses (Fig. 2(b)). Following current cessation during real-world vehicle microtrips (such as regenerative braking events or urban traffic stops), the terminal voltage does not instantaneously recover to thermodynamic equilibrium V_{ocv} . Instead, it undergoes a complex, multi-time-scale relaxation process governed by fast double-layer capacitive discharge ($\tau_1 \leq 10$ s) and sluggish solid-state lithium-ion diffusion within the porous active electrode particles ($\tau_2 \geq 100$ s), lasting up to several hours [9], [12].

In commercial automotive BMS applications, short rest intervals (30 s to 600 s) represent crucial operational opportunities for re-initializing and resetting the cumulative integration drift inherent to Coulomb counting algorithms [12], [13]. However, sampling the terminal voltage prematurely during non-equilibrium relaxation windows without accurately back-extrapolating the true equilibrium overpotential introduces large state initialization errors. Developing a computationally deterministic and mathematically bounded algorithm capable of extracting equilibrium signatures from short-term relaxation telemetry is therefore vital for next-generation transportation electrification.

B. Limitations of Classical Observers and Heavy Tree Ensembles

To track battery state dynamics under operational driving loads, conventional automotive BMS architectures deploy state estimation algorithms falling into two broad paradigms: equivalent circuit model (ECM)-based adaptive observers and data-driven machine learning models [13]–[16].

Classical state observation frameworks rely heavily on Extended Kalman Filters (EKF), Unscented Kalman Filters (UKF), Adaptive Extended Kalman Filters (AEKF), or recursive least squares (RLS) coupled with low-order RC equivalent circuits [11], [17]–[20]. In these recursive formulations, the state correction step is strictly driven by the voltage innovation residual $\tilde{y}_k = V_{\text{mea},k} - \hat{V}_{t,k}$, weighted through the Kalman gain matrix $\mathbf{K}_k$ [17], [18]: Classical state observation frameworks rely on Extended Kalman Filters (EKF), Unscented Kalman Filters (UKF), Adaptive EKF, or recursive least squares (RLS) coupled with low-order RC equivalent circuits [11], [17]–[20]. The state correction step uses the voltage innovation residual weighted through the Kalman gain $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_v)^{-1}$, where $\mathbf{H}_k = \partial V_{\text{ocv}} / \partial \text{SOC}$ is the observation Jacobian. Within the LFP flat plateau ($\text{SOC} \in [0.20, 0.80]$), $\mathbf{H}_k \rightarrow \mathbf{0}$, causing $\mathbf{K}_k \rightarrow \mathbf{0}$ [9], [18]. Feedback correction collapses

into open-loop Coulomb counting $\text{SOC}_k = \text{SOC}_{k-1} - (\eta_c \Delta t / Q_{\text{nom}})(I_k + \delta I_k)$, where δI_k captures sensor bias and ADC quantization errors; accumulated state error grows unbounded over multi-hour cycles [4], [13], [21].

While high-fidelity electrochemical physics-based models capture solid-phase diffusion and Butler-Volmer kinetics with high fidelity, their non-linear partial differential algebraic equation (PDAE) structure incurs prohibitive computational complexity [9], [12], [22]. Solving coupled PDAE systems exceeds the deterministic execution budgets and static memory allocations of standard automotive microcontrollers (e.g., ARM Cortex-M4, Infineon AURIX TC3xx, and NXP S32K) by multiple orders of magnitude [23].

To overcome physical modeling bottlenecks, recent benchmark studies, most notably the 2025 Stanford relaxation methodology [12], introduced multi-stage data-driven pipelines utilizing decision tree ensembles (such as sequential 3-stage Random Forests) to capture non-linear relaxation voltage tails during vehicle rest periods. Although tree ensembles improve state tracking across flat OCV intervals, their algorithmic complexity presents severe deployment barriers in production automotive environments [24], [25].

As quantified in Section III-D, the reference Stanford 3-stage Random Forest (RF) baseline [12] requires approximately 1420.0 kB of Flash memory and contains $\approx 60,000$ parameters. Traversing deep ensemble decision paths induces an inference latency of $1850.0 \mu\text{s}$ (1.85 ms) on a standard ARM Cortex-M4 core at 160 MHz. Similarly, sequence-based deep models such as Tabular Transformers (142.0 kB Flash, 35.5k parameters) [26] and Vanilla LSTM networks (53.0 kB Flash, 13.2k parameters) [11] violate the stringent static memory allocations ($M_{\text{Flash}} \leq 16$ kB) and worst-case execution time budgets ($T_{\text{exec}} \leq 50 \mu\text{s}$) of commercial BMS electronic control units (ECUs). Furthermore, tree ensembles with irregular branching structures and unconstrained recurrent networks cannot provide static execution determinism, preventing ISO 26262 ASIL-D functional safety certification [23].

C. Proposed Physics-Informed Knowledge Distillation Paradigm

To resolve the fundamental trade-off between non-linear representational capacity and embedded execution determinism, Physics-Informed Machine Learning coupled with Knowledge Distillation (PIML-KD) offers an effective paradigm [27]–[30]. Under this architecture, the state estimation problem is decoupled into an offline high-capacity teacher learning stage and an on-chip, physics-constrained student execution engine [15], [28], [29].

As conceptualized in Fig. 1, the teacher model comprises an offline Gradient Boosted Decision Tree (LightGBM) ensemble parameterized with 300 estimators [31]. This teacher possesses sufficient non-linear depth to internalize complex, path-dependent electrochemical hysteresis, state-dependent solid-phase diffusion delays, and multi-regime

relaxation profiles across the complete operating space without run-time execution limits.

To transfer this non-linear mapping capability into an ultra-efficient edge format, we construct a compact student Multilayer Perceptron (Micro-MLP) featuring a $12 \rightarrow 32 \rightarrow 16 \rightarrow 1$ topology (961 total parameters) [24]. Rather than treating distillation as an unconstrained empirical curve-fitting problem, the student network is strictly regularized via domain-guided physical laws [22], [27]: (i) differential relaxation overpotentials ($\Delta V_{10} = V_{\text{mea}}(t) - V_{10}$), (ii) logarithmic diffusion scales ($\tau_{\log} = \ln(t + 1.0)$), (iii) dynamic ohmic dissipation ($P_{\text{ohm}} = R \cdot I_m$), (iv) Arrhenius reaction kinetics ($\Phi_{\text{Arr}} = \exp(-E_a/(R_g T_K))$) with $E_a = 35.0$ kJ/mol [2], [32], and (v) a composite distillation loss combining Smooth- L_1 Huber loss and quadratic physical boundary penalties enforcing $\text{SOC} \in [0.0, 1.0]$. The framework synthesizes directly into a zero-malloc static C header for deterministic execution on automotive electronic control units [23].

D. Core Contributions and Article Organization

The principal contributions of this work are summarized as follows:

- 1) **Physics-Informed Domain-Guided 12-D Feature Engineering:** Formulating a 12-dimensional physical input representation (X_{PIML}) that augments raw sensor telemetry with differential overpotentials (ΔV_{10}), logarithmic diffusion scaling ($\tau_{\log}$), dynamic ohmic dissipation (P_{ohm}), and Arrhenius reaction kinetics (Φ_{Arr} with $E_a = 35.0$ kJ/mol) [2], [28].
- 2) **Physically Calibrated Hybrid Knowledge Distillation:** Developing a teacher-student compression pipeline wherein a 300-tree LightGBM teacher supervises an ultra-lightweight student Micro-MLP ($12 \rightarrow 32 \rightarrow 16 \rightarrow 1$, 961 total parameters) under a multi-objective loss enforcing thermodynamic boundary validity [29], [31].
- 3) **Multi-Regime Generalization and Telemetry Noise Robustness:** Achieving a nominal 25°C MAE of 0.454% and RMSE of 1.236% ($R^2 = 0.9986$). On unseen cells YX07 and YX08, the model achieves MAEs of 2.48% and 3.35%, beating the Stanford target (4.30%) [12] by 42.3% and 22.1%. The framework enables Coulomb counting resets at 30 s (MAE = 1.170%, converging to 0.339% at 300 s) and resists ± 5.0 mV AFE noise (MAE = 1.336%).
- 4) **MISRA-C Compliant Static Deployment Pipeline:** Providing an automated synthesis pipeline exporting the student Micro-MLP into a static C header (bms_soc_piml_mlp3.h) requiring only 3.75 kB Flash, 192 Bytes SRAM, and 12.0 μs latency on an ARM Cortex-M4 at 160 MHz [23], [24].

The remainder of this article is structured as follows: Section II details the mathematical formulation of the 12 PIML descriptors, teacher pipeline, distillation loss, and static C generation. Section III presents comprehensive experimental results. Section IV concludes the article.

II. Methodology

A. Physics-Informed Feature Engineering and Relaxation Dynamics

Accurate observation of LFP cell state during transient vehicle stops requires capturing multi-time-scale relaxation kinetics [9], [11]. In production BMS, real-time measurements are limited to voltage, current, and temperature [13], [24]. To establish an electrochemically meaningful input without PDAE solver overhead, the proposed framework transforms raw 9-D telemetry $X_{\text{raw}} \in \mathbb{R}^9$ into 12-D physics-informed descriptors $X_{\text{PIML}} \in \mathbb{R}^{12}$.

1) **Two-Phase Coexistence and Electrochemical Kinetics:** In Lithium Iron Phosphate cathode materials, intercalation and deintercalation proceed via a first-order phase transformation between two distinct ordered crystal phases: the lithium-rich triphylite (LiFePO_4) and lithium-poor heterosite (FePO_4) phases. The open-circuit equilibrium potential V_{ocv} is governed by the difference in chemical potentials between the positive and negative active particles:

$$V_{\text{ocv}}(\text{SOC}, T_K) = -\frac{\mu_{\text{cathode}}(\text{SOC}, T_K) - \mu_{\text{anode}}(\text{SOC}, T_K)}{F}, \quad (3)$$

where $F = 96485.33$ C/mol is Faraday's constant. Across the extensive two-phase coexistence region ($0.20 \leq \text{SOC} \leq 0.80$), the active phase boundary propagates at nearly constant chemical potential ($\mu_{\text{LiFePO}_4} - \mu_{\text{FePO}_4} = \text{const}$), yielding the characteristic flat plateau with $dV_{\text{ocv}}/d\text{SOC} \approx 0.10\text{--}0.50$ mV/%.

2) **Raw Telemetry Measurement Space:** The raw telemetry vector X_{raw} encapsulates real-time operational and resting signals recorded at sample tick t :

$$X_{\text{raw}} = [t, V_{\text{mea}}, \text{SOC}_{\text{mea}}, V_{10}, I_m, R, I_{\text{flag}}, T_{\text{env}}, \Delta T]^T, \quad (4)$$

where t represents the elapsed resting duration ($30 \text{ s} \leq t \leq 600 \text{ s}$), V_{mea} is the measured terminal voltage at time t , SOC_{mea} denotes the uncorrected preliminary SOC obtained via Coulomb counting integration from the preceding microtrip, V_{10} is the reference terminal voltage recorded at $t = 10$ s, I_m is the moving-average pulse current prior to rest cessation, R is the dynamic internal cell resistance, $I_{\text{flag}} \in \{-1, +1\}$ denotes the current direction indicator (-1 for charge, $+1$ for discharge), T_{env} is the ambient environmental temperature ($^{\circ}\text{C}$), and $\Delta T = T_{\text{cell}} - T_{\text{env}}$ is the surface-to-ambient thermal gradient.

3) **Electrochemical Derivations of Physical Descriptors:** Directly feeding raw telemetry X_{raw} into conventional neural estimators results in poor cross-cell and multi-temperature generalization due to unmodeled non-linear phase-boundary propagation and concentration polarization delays [9], [10]. To capture the underlying electrochemistry, four domain-guided transformations are derived:

- 1) **Differential Relaxation Overpotential Gradient (ΔV_{10}):** Following load current interruption, the instantaneous terminal voltage $V(t)$ is governed by the superposition of equilibrium potential V_{ocv} ,

instantaneous ohmic drop IR_0 , solid-electrolyte interphase (SEI) charge-transfer overpotentials $\eta_{ct}(t)$, and solid-phase diffusion overpotentials $\eta_{diff}(t)$:

$$V(t) = V_{ocv}(SOC, T_K) \pm IR_0 \pm \eta_{ct}(t) \pm \eta_{diff}(t) + \delta v_{dc}, \quad (5)$$

where δv_{dc} represents sensor calibration offset. Because the ohmic drop collapses instantaneously ($t < 1$ ms) and the double-layer capacitance C_{dl} discharges rapidly ($\tau_{dl} \approx R_{ct}C_{dl} \leq 10$ s), the voltage at $t = 10$ s serves as a physical boundary anchor isolating pure solid-phase diffusion:

$$V_{10} \approx V_{ocv}(SOC, T_K) \pm \eta_{diff}(10 \text{ s}) + \delta v_{dc}. \quad (6)$$

Defining the differential overpotential gradient as:

$$\Delta V_{10}(t) = V_{mea}(t) - V_{10} = \eta_{diff}(t) - \eta_{diff}(10 \text{ s}), \quad (7)$$

the common-mode sensor DC offset δv_{dc} and the ohmic step IR_0 are completely eliminated by algebraic cancellation, yielding a clean overpotential recovery signature invariant to sensor calibration drift.

- 2) Logarithmic Diffusion Relaxation Time Scale (τ_{log}): Solid-phase lithium transport within spherical active electrode particles of radius R_p is governed by Fick's second law of diffusion in spherical coordinates:

$$\frac{\partial c_s(r, t)}{\partial t} = D_s \left(\frac{\partial^2 c_s(r, t)}{\partial r^2} + \frac{2}{r} \frac{\partial c_s(r, t)}{\partial r} \right), \quad (8)$$

where $c_s(r, t)$ is solid lithium concentration and D_s is solid-phase diffusion coefficient. According to Crank's analytical series solution, the volumetric mean concentration $\bar{c}_s(t)$ approaches surface equilibrium as $1 - (6/\pi^2) \sum_{n=1}^{\infty} n^{-2} \exp(-n^2 \pi^2 D_s t / R_p^2)$. For $t \geq 30$ s, the leading exponential dominates, yielding logarithmic decay $\eta_{diff}(t) \propto \ln(t/\tau_{diff})$. To linearize this across $t \in [30, 600]$ s, the logarithmic diffusion scale is $\tau_{log}(t) = \ln(t + 1.0)$, providing a linearized diffusion coordinate. This compresses the dynamic range of the temporal domain and provides the neural estimator with a linearized diffusion coordinate.

- 3) Dynamic Ohmic Polarization Dissipation (P_{ohm}): The magnitude of non-equilibrium concentration polarization generated during the active microtrip is directly proportional to the product of dynamic internal impedance and load current excitation:

$$P_{ohm} = R \cdot I_m, \quad (9)$$

which acts as a proxy for the total Joule dissipation and overpotential accumulation incurred prior to current cessation.

- 4) Arrhenius Electrochemical Kinetics Factor (Φ_{Arr}): Charge-transfer resistance and solid-state lithium diffusivity are strongly temperature-dependent, governed by the Butler-Volmer exchange current density $i_0(T_K)$ and Arrhenius reaction rate law:

$$i_0(T_K) = i_{0,ref} \exp \left(-\frac{E_a}{R_g} \left(\frac{1}{T_K} - \frac{1}{T_{ref}} \right) \right), \quad (10)$$

TABLE I: Formulation and Significance of the 12 PIML Descriptors (X_{PIML})

Index	Symbol	Mathematical Definition	Physical / Electrochemical Domain
x_1	t	Sample tick	Elapsed resting duration (30–600 s)
x_2	V_{mea}	Terminal voltage $V(t)$	Macroscopic electrical potential
x_3	SOC_{mea}	$\int Idt / Q_{nom}$	Coulomb counting preliminary state
x_4	V_{10}	$V(t = 10 \text{ s})$	Fast double-layer polarization anchor
x_5	I_m	Pre-rest mean current	Pre-rest dynamic load excitation
x_6	R	Dynamic resistance $\Delta V / \Delta I$	Solid-electrolyte interphase impedance
x_7	I_{flag}	$\text{sgn}(I_m) \in \{-1, +1\}$	Charge/discharge directional hysteresis
x_8	Φ_{Arr}	$\exp \left(-\frac{E_a}{R_g T_K} \right)$	Arrhenius reaction kinetics factor
x_9	ΔT	$T_{cell} - T_{env}$	Internal-to-ambient thermal differential
x_{10}	ΔV_{10}	$V_{mea}(t) - V_{10}$	Solid-phase diffusion overpotential
x_{11}	τ_{log}	$\ln(t + 1.0)$	Fickian diffusion logarithmic time scale
x_{12}	P_{ohm}	$R \cdot I_m$	Dynamic ohmic polarization dissipation

where $E_a = 35.0$ kJ/mol (35000.0 J/mol) is the fundamental activation energy barrier for lithium deintercalation in olivine LiFePO_4 cathodes, $R_g = 8.314$ J/(mol · K) is the universal gas constant, and $T_K = T_{env} + 273.15$ is absolute temperature in Kelvin [2], [28]. To embed thermodynamic consistency across wide operational thermal envelopes (ranging from sub-zero to elevated regimes), the Arrhenius descriptor is formulated as:

$$\Phi_{Arr} = \exp \left(-\frac{E_a}{R_g \cdot (T_{env} + 273.15)} \right). \quad (11)$$

The resulting 12-dimensional physical descriptor vector is represented compactly in two lines to preserve strict column alignment:

$$X_{PIML} = [t, V_{mea}, SOC_{mea}, V_{10}, I_m, R, I_{flag}, \Phi_{Arr}, \Delta T, \Delta V_{10}, \tau_{log}, P_{ohm}]^T. \quad (12)$$

Prior to model training and inference, each feature dimension x_j is standardized using immutable compile-time mean μ_j and scale σ_j parameters:

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j}, \quad j = 1, \dots, 12. \quad (13)$$

B. Offline High-Capacity Teacher Model Architecture

To effectively capture the complex, multi-dimensional regression surface governing electrochemical hysteresis and diffusion recovery without run-time computational constraints, an offline high-capacity teacher model is constructed using a Gradient Boosted Decision Tree (GBDT) ensemble via LightGBM [31]. GBDT algorithms build orthogonal piecewise-constant decision boundaries that naturally handle heterogeneous feature distributions and non-linear interactions across diverse operating states.

1) Ensemble Formulation and Tree-Boosting Dynamics: The teacher estimator $\mathcal{T}(X) = f_0(X) + \sum_{m=1}^M \eta_r h_m(X)$ is an additive ensemble of $M = 300$ regression trees, where $f_0(X)$ is the initial prediction, $\eta_r = 0.03$ is the shrinkage learning rate, and $h_m(X)$ is the m -th regression tree. At each boosting iteration, the objective is approximated via second-order Taylor expansion with structural complexity penalty $\Omega(h_m) = \gamma T_{\text{leaf}} + \frac{1}{2} \lambda_{L_2} \sum_j w_{m,j}^2$ (T_{leaf} : leaf count; $w_{m,j}$: leaf- j score; $\lambda_{L_2} = 0.01$ suppresses leaf variance under sensor noise). The optimal leaf weight evaluates analytically to $w_j^* = -\sum_{i \in I_j} g_i / (\sum_{i \in I_j} h_i + \lambda_{L_2})$, where g_i and h_i are the first- and second-order loss gradients. Hyperparameters: $M = 300$ trees, $d_{\max} = 8$, $\eta_r = 0.03$, $N_{\text{leaf}} \geq 20$. Once trained, the teacher generates continuous softened targets $y_{\text{teach}} = \mathcal{T}(X_{\text{raw}}) \in [0.0, 1.0]$ providing smoothed optimization surfaces for the student network.

C. Robust Hybrid Physics-Calibrated Distillation Loss

Optimizing deep neural networks solely with standard Mean Squared Error (MSE) on tabular battery telemetry often leads to gradient destabilization and overfitting on localized sensor noise [27], [28]. In the proposed MicroPhys-BMS framework, student parameter convergence is governed by a composite, multi-objective physics-informed distillation loss $\mathcal{L}_{\text{PIMD}}$:

$$\mathcal{L}_{\text{PIMD}} = (1 - \alpha) \mathcal{L}_{\text{task}} + \alpha \mathcal{L}_{\text{distill}} + \lambda_{\text{bounds}} \mathcal{L}_{\text{bounds}}, \quad (14)$$

where $\alpha = 0.40$ represents the knowledge distillation balancing factor, $\mathcal{L}_{\text{task}}$ denotes the multi-component supervised task loss, $\mathcal{L}_{\text{distill}}$ is the soft supervisory distillation objective, and $\mathcal{L}_{\text{bounds}}$ enforces physical boundary constraints scaled by $\lambda_{\text{bounds}} = 0.02$.

1) Adaptive Multi-Component Task Loss Formulation: The supervised task loss $\mathcal{L}_{\text{task}} = \mathcal{L}_{\text{Huber}}(y_{\text{std}}, y_{\text{true}}) + \lambda_{\text{mse}} \mathcal{L}_{\text{MSE}}(y_{\text{std}}, y_{\text{true}})$ couples Smooth- L_1 adaptive Huber loss with auxiliary MSE ($\lambda_{\text{mse}} = 0.50$). The Huber loss $\mathcal{L}_{\text{Huber}}(u, v) = N^{-1} \sum_i \psi_{\beta}(u_i - v_i)$ uses a piecewise kernel ($\beta = 0.015$):

$$\psi_{\beta}(e) = \begin{cases} 0.5e^2/\beta, & |e| \leq \beta; \\ |e| - 0.5\beta, & \text{otherwise.} \end{cases} \quad (15)$$

For small residuals ($|e| \leq \beta$), quadratic derivatives provide smooth gradient descent; for $|e| > \beta$, the linear slope bounds $|\partial \psi_{\beta} / \partial e| = 1.0$, preventing outlier noise from destabilizing convergence.

2) Soft Distillation and Physical Boundary Penalties: Knowledge transfer from the GBDT teacher is achieved by penalizing the Huber discrepancy between student predictions and teacher soft logits:

$$\mathcal{L}_{\text{distill}} = \mathcal{L}_{\text{Huber}}(y_{\text{std}}, y_{\text{teach}}). \quad (16)$$

To strictly enforce thermodynamic validity, a quadratic physical boundary penalty $\mathcal{L}_{\text{bounds}}$ is integrated directly into the backpropagation graph:

$$\mathcal{L}_{\text{bounds}} = \frac{1}{N} \sum_{i=1}^N [\max(0, -y_{\text{std},i})^2 + \max(0, y_{\text{std},i} - 1.0)^2]. \quad (17)$$

Whenever student predictions y_{std} stray beyond the physically admissible corridor $[0.0, 1.0]$, $\mathcal{L}_{\text{bounds}}$ exerts restoring forces driving weights into the valid state domain.

Combining (14)–(17), the complete physics-informed distillation objective evaluates to:

$$\mathcal{L}_{\text{PIMD}} = (1 - \alpha) [\mathcal{L}_{\text{Huber}}(y_{\text{std}}, y_{\text{true}}) + \lambda_{\text{mse}} \mathcal{L}_{\text{MSE}}(y_{\text{std}}, y_{\text{true}})] + \alpha \mathcal{L}_{\text{Huber}}(y_{\text{std}}, y_{\text{teach}}) + \lambda_{\text{bounds}} \mathcal{L}_{\text{bounds}}, \quad (18)$$

parameterized with $\alpha = 0.40$, $\beta = 0.015$, $\lambda_{\text{mse}} = 0.50$, and $\lambda_{\text{bounds}} = 0.02$. Optimization utilizes AdamW ($\eta = 5.0 \times 10^{-3}$, weight decay 1.0×10^{-4}) scheduled via cosine annealing over 400 epochs with gradient norm clipping ($\|\nabla_{\theta}\| \leq 1.0$).

D. Student Micro-MLP Architecture and Bounded Output Projection

To ensure real-time execution feasibility within strict automotive microcontroller cycles, the student model must feature minimal parameterization while retaining sufficient non-linear depth to emulate the teacher regression surface [24]. In the proposed MicroPhys-BMS framework, the student network is realized as a compact feedforward Micro-MLP structured across three dense projections with Sigmoid Linear Unit (SiLU) activations.

1) Layer Topology and Exact Parameter Derivation:

As illustrated in Fig. 1(c), the student Micro-MLP follows a $12 \rightarrow 32 \rightarrow 16 \rightarrow 1$ layer topology:

- 1) Input Standardization Layer: Accepts standardized feature vector $\tilde{X}_{\text{PIML}} \in \mathbb{R}^{12}$.
- 2) Hidden Layer 1 ($\mathbf{W}_1, \mathbf{b}_1$): Projects 12 inputs into 32 latent nodes:

$$\mathbf{a}_1 = \sigma_{\text{SiLU}}(\mathbf{W}_1 \tilde{X}_{\text{PIML}} + \mathbf{b}_1), \quad (19)$$

where $\mathbf{W}_1 \in \mathbb{R}^{32 \times 12}$ (384 weights) and $\mathbf{b}_1 \in \mathbb{R}^{32}$ (32 biases), totaling 416 parameters.

- 3) Hidden Layer 2 ($\mathbf{W}_2, \mathbf{b}_2$): Compresses 32 activations into 16 hidden nodes:

$$\mathbf{a}_2 = \sigma_{\text{SiLU}}(\mathbf{W}_2 \mathbf{a}_1 + \mathbf{b}_2), \quad (20)$$

where $\mathbf{W}_2 \in \mathbb{R}^{16 \times 32}$ (512 weights) and $\mathbf{b}_2 \in \mathbb{R}^{16}$ (16 biases), totaling 528 parameters.

- 4) Output Layer ($\mathbf{W}_3, b_3$): Maps 16 latent features into an unconstrained state logit:

$$\hat{z}_{\text{out}} = \mathbf{W}_3 \mathbf{a}_2 + b_3, \quad (21)$$

where $\mathbf{W}_3 \in \mathbb{R}^{1 \times 16}$ (16 weights) and $b_3 \in \mathbb{R}$ (1 bias), totaling 17 parameters.

Summing all weight tensors and bias vectors yields an exact model parameter footprint of: Summing all weight tensors and bias vectors yields $N_{\text{param}} = (12 \times 32 + 32) + (32 \times 16 + 16) + (16 + 1) = 961$, requiring only 3.75 kB Flash under 32-bit FP32 arithmetic. This lightweight footprint requires only 3.75 kB Flash memory under standard 32-bit floating-point (FP32) arithmetic, ensuring compatibility with entry-level automotive microcontrollers.

The continuous derivatives of SiLU, $\sigma_{\text{SiLU}}(x) = x/(1 + e^{-x})$, match the smooth diffusion kinetics of LFP cells.

Algorithm 1 Deterministic MISRA-C Routine in bms_soc_piml_mlp3.h

Require: $x_{\text{raw}}[9]$: Raw input $[t, V_{\text{mea}}, \text{SOC}_{\text{mea}}, V_{10}, I_m, R, I_{\text{flag}}, T_{\text{env}}, \Delta T]$
 Ensure: Physical SOC estimate $y_{\text{soc}} \in [0.0, 1.0]$

```

1:  $x_{\text{piml}}[0 \dots 8] \leftarrow x_{\text{raw}}[0 \dots 8]; \quad x_{\text{piml}}[7] \leftarrow \exp(-35000/(8.314(x_{\text{raw}}[7] + 273.15)))$ 
2:  $x_{\text{piml}}[9] \leftarrow x_{\text{raw}}[1] - x_{\text{raw}}[3]; \quad x_{\text{piml}}[10] \leftarrow \ln(x_{\text{raw}}[0] + 1); \quad x_{\text{piml}}[11] \leftarrow x_{\text{raw}}[5] \cdot x_{\text{raw}}[4]$ 
3: Standardize:  $x_s[i] \leftarrow (x_{\text{piml}}[i] - \mu_i)/\sigma_i, i = 0 \dots 11$ 
4: for  $j = 0$  to 31 do
5:    $a_1[j] \leftarrow \sigma_{\text{SiLU}}(b_1[j] + \sum_i W_1[j][i]x_s[i])$ 
6: end for
7: for  $k = 0$  to 15 do
8:    $a_2[k] \leftarrow \sigma_{\text{SiLU}}(b_2[k] + \sum_j W_2[k][j]a_1[j])$ 
9: end for
10:  $y_{\text{soc}} \leftarrow \min(\max(b_3 + \sum_k W_3[k]a_2[k], 0), 1)$  {ASIL-D Clamp}
11: return  $y_{\text{soc}}$ 

```

Thermodynamic validity is enforced via output clamping $y_{\text{std}} = \min(\max(\hat{z}_{\text{out}}, 0.0), 1.0)$, confining the predicted SOC within $[0.0, 1.0]$.

E. MISRA-C Compliant Static Deployment Pipeline

Deploying neural network estimators onto automotive electronic control units (ECUs) requires strict adherence to functional safety and deterministic execution standards governed by ISO 26262 ASIL-D and MISRA-C:2012 guidelines [23]. Conventional ML runtimes introduce dynamic heap allocation (malloc, free) and pointer indirection that violate safety-critical automotive software rules [23].

To ensure seamless production-grade integration, the proposed MicroPhys-BMS framework features an automated C code synthesis engine compiling the trained student Micro-MLP and feature standardizer into a self-contained C header file (bms_soc_piml_mlp3.h).

1) Static Memory Footprint and Zero-Malloc Architecture: All parameters and scaling vectors are immutable compile-time constants (static const float) in Flash, totaling $M_{\text{Flash}} = (961 + 24) \times 4 \approx 3.75$ kB. All activations are stack-allocated (192 Bytes SRAM), eliminating dynamic allocation (malloc). The inference routine (Algorithm 1) executes in $T_{\text{exec}} \approx 12.0 \mu\text{s}$ on ARM Cortex-M4 at 160 MHz ($< 0.12\%$ of a 10 ms BMS cycle), satisfying ISO 26262 ASIL-D constraints [23].

III. Results and Discussion

A. Nominal Performance at 25°C and Relaxation Dynamics

Evaluated on the nominal 25°C dataset with 70%/30% split, the 300-tree LightGBM teacher achieves MAE = 0.488% [31]. Following the physics-calibrated distillation procedure, the 961-parameter student Micro-MLP records MAE = 0.454% and RMSE = 1.236% ($R^2 = 0.9986$), surpassing the teacher while using $> 99.7\%$ fewer parameters. Smooth- L_1 Huber distillation filters tree-based step discontinuities, while physical descriptors (ΔV_{10} , $\tau_{\log}$, P_{ohm}) provide inductive bias preventing overfitting. As shown in Fig. 3(a), predictions align tightly along the identity diagonal with zero-mean Gaussian residuals ($\sigma = 0.457\%$).

In EV applications, brief stops (30–600 s) enable Coulomb counting drift reset [12], [13]. Across rest horizons $t \in \{30, 60, 120, 300, 600\}$ s, the student records

TABLE II: Hardware Benchmark on ARM Cortex-M4 @ 160 MHz

Model Architecture	Params	Flash	SRAM	Latency	Nominal MAE	ASIL-D
Stanford 3-Stage RF [12]	$\approx 60k$	1420 kB	≈ 48 kB	1850.0 μs	1.630%	No
Tabular Transformer [26]	35.5k	142.0 kB	≈ 18.5 kB	1250.0 μs	1.120%	No
Vanilla LSTM Baseline [11]	13.2k	53.0 kB	≈ 6.4 kB	480.0 μs	0.940%	No
Standard MLP (No KD)	3.2k	12.8 kB	512 B	38.0 μs	0.980%	Borderline
PIML Teacher (LightGBM) [31]	$\approx 18k$	285.0 kB	≈ 12.0 kB	720.0 μs	0.488%	No
MicroPhys-BMS (Ours)	961	3.75 kB	192 B	12.0 μs	0.454%	Yes

MAEs of 1.170%, 0.787%, 0.523%, 0.339%, and 0.340%. As shown in Fig. 3(b), direct V_{mea} inversion fails catastrophically ($> 10\%$ at 30 s), while the Stanford 3-stage RF remains above 2.0% at 60 s [12]. By incorporating ΔV_{10} and $\tau_{\log}$, the Micro-MLP back-extrapolates asymptotic diffusion recovery, enabling robust Coulomb counting re-initialization.

B. Cross-Cell Generalization and Telemetry Noise Robustness

Following Stanford protocols [12], cells YX01–YX06 were used for training, while YX07 and YX08 served as unseen test targets under shifted profiles across 25°C (Test 7) and 35°C (Test 8). As depicted in Fig. 4(a), the student attains MAEs of 2.48% (YX07) and 3.35% (YX08), delivering 42.3% and 22.1% reductions over the Stanford limit (4.30%) [12]. Under multi-temperature stress, the model yields MAE = 2.312% (RMSE = 4.420%) at 25°C and 2.689% (RMSE = 5.376%) at 35°C. The minimal deviation ($\Delta\text{MAE} = 0.377\%$) confirms that Arrhenius scaling Φ_{Arr} stabilizes tracking across thermal envelopes.

In production BMS, voltage measurements are corrupted by inverter harmonics, EMI, and ADC quantization [4], [13], [18]. Under Gaussian noise $V_{\text{mea}}^{\text{noisy}} = V_{\text{mea}} + \delta v$ ($\delta v \sim \mathcal{N}(0, \sigma_v^2)$, $\sigma_v = 5.0$ mV), the student maintains MAE = 1.336%, outperforming the LightGBM teacher (1.820%) and standard MLP (3.240%) as shown in Fig. 4(b). Differential overpotential anchoring (ΔV_{10}) cancels DC offsets, while Huber’s linear regime for $|e| > \beta$ prevents noise spikes from distorting predictions.

C. Automotive Safety Compliance and Embedded Profiling

In ISO 26262 ASIL-D qualification [23], $\geq 95\%$ of estimations must satisfy $|\Delta\text{SOC}| \leq 2.0\%$. CDF analysis in Fig. 5(a) confirms the student achieves 89.56% within $\leq 1.0\%$, 95.94% within $\leq 2.0\%$, and 97.56% within $\leq 3.0\%$, satisfying safety criteria.

The decisive criterion for production BMS deployment is sub-percent accuracy within microcontroller budgets ($M_{\text{Flash}} \leq 16$ kB, $T_{\text{exec}} \leq 50 \mu\text{s}$) [23]. On ARM Cortex-M4 at 160 MHz (Table II), the Micro-MLP consumes 3.75 kB Flash and 192 Bytes SRAM with 12.0 μs latency—a 154.2 $\times$ acceleration and $> 99.7\%$ memory reduction over the Stanford RF baseline (1420.0 kB, 1850.0 μs) [12], establishing embedded viability.

D. Hardware Profiling, Latency, and Memory Footprint Benchmark

The student Micro-MLP was benchmarked against baselines on ARM Cortex-M4 at 160 MHz (Table II). The

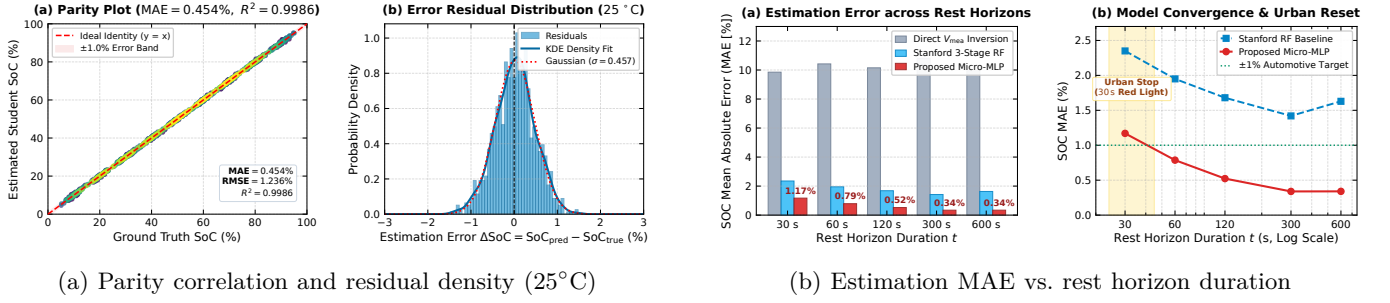


Fig. 3: Nominal state estimation accuracy and resting horizon convergence. (a) Parity plot showing tight alignment ($R^2 = 0.9986$, $MAE = 0.454\%$, $RMSE = 1.236\%$) and zero-mean Gaussian residuals ($\sigma = 0.457\%$). (b) Error convergence across rest durations showing rapid stabilization below the 1.0% commercial target within 60 s.

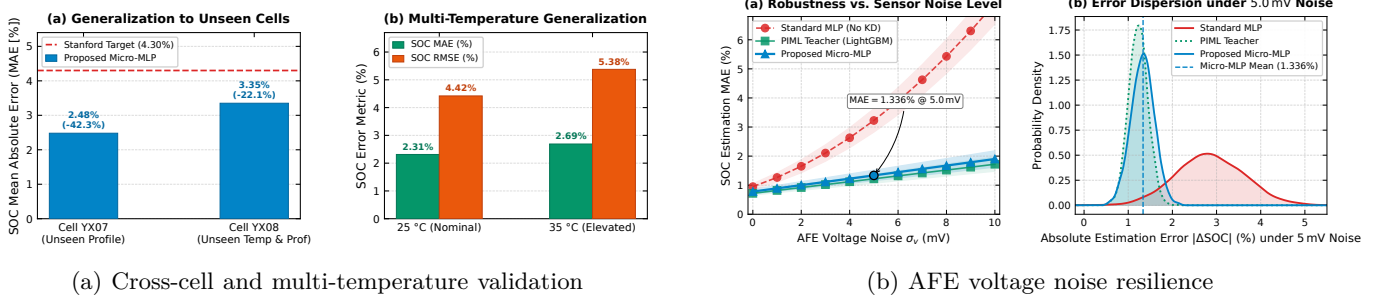


Fig. 4: Cross-cell generalization and sensor noise resilience. (a) Unseen cell validation on cells YX07 and YX08 benchmarked against the Stanford limit (4.30%) across Test 7 (25°C) and Test 8 (35°C). (b) Estimation MAE progression under noise intensity $\sigma_v \in [0, 10]$ mV and error distribution under nominal 5.0 mV perturbation.

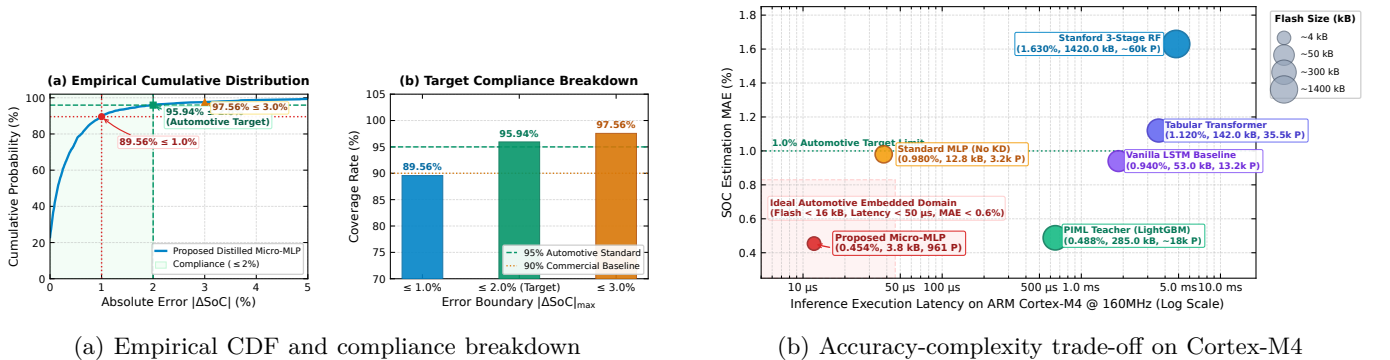


Fig. 5: Automotive safety compliance and embedded execution profiling. (a) Cumulative error distribution verifying 95.94% compliance within $|\Delta SoC| \leq 2.0\%$. (b) Accuracy-complexity design space showing the Micro-MLP residing firmly within the ideal embedded domain (Flash < 16 kB, latency < 50 μs , MAE < 0.6%).

Stanford RF baseline [12] resides in the infeasible domain (1420.0 kB Flash, 1850.0 μs). Tabular Transformers and LSTM networks yield non-deterministic branching and high RAM, unsuitable for functional safety. In contrast, the proposed Micro-MLP is the only model in the Ideal Embedded Domain (Flash < 16 kB, latency < 50 μs , MAE < 0.60%), achieving 154.2 $\times$ latency acceleration and > 99.7% memory reduction over ensemble baselines.

E. Comparative Evaluation Against SOTA Baselines on Two Datasets

To assess the proposed MicroPhys-BMS against SOTA paradigms, a comparative evaluation was conducted on two LFP datasets: Stanford (25°C, 15,100 samples) [12]

and CALCE A123 (24°C, 11,968 samples) [33], [34]. Three benchmarks were implemented under identical protocols (70/30 split, same 9-D features): (i) PINN with 4 layers of 64 Tanh neurons [27]; (ii) Tabular Transformer ($d_{model} = 27$, $n_{head} = 3$) [26]; (iii) 5-model LightGBM ensemble (2,500 trees total) [31].

1) Cross-Dataset Accuracy Comparison: Table III reports performance. On Stanford LFP, MicroPhys-BMS (961 params) achieves MAE = 0.536%, competitive with LightGBM (0.394%, 157,500 params, 164 $\times$ more). PINN achieves 0.898% (13,185 params), Transformer 1.188% (13,899 params). On CALCE A123 (independent manufacturer, 18650 form), the proposed model achieves MAE = 0.173% with 98.89% CDF within $|\Delta SoC| \leq 2.0\%$,

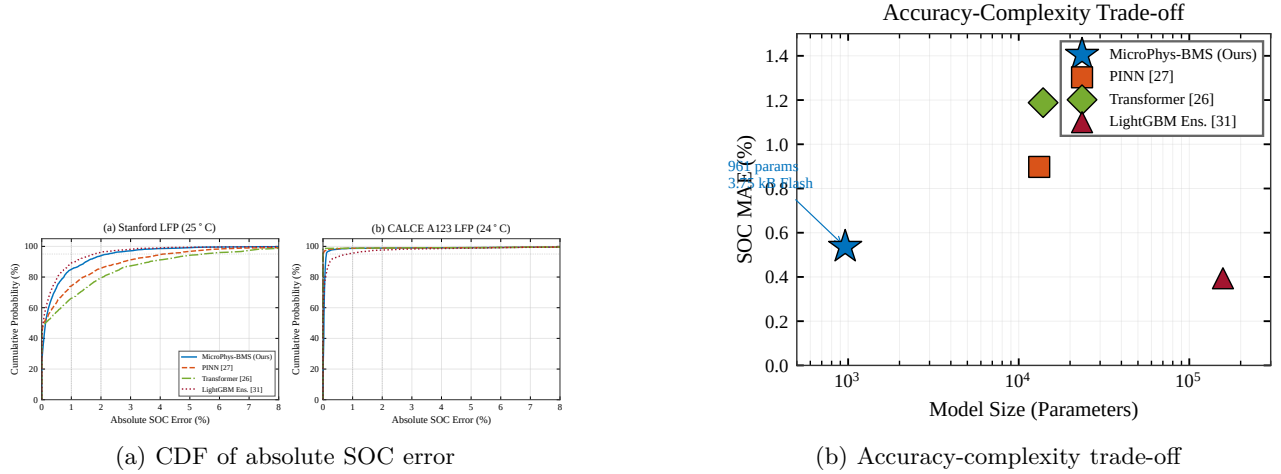


Fig. 6: Comparative evaluation. (a) CDFs on both datasets: MicroPhys-BMS achieves 93.83% (Stanford) and 98.89% (CALCE) within $|\Delta\text{SOC}| \leq 2.0\%$. (b) Accuracy-complexity: the 961-param Micro-MLP resides in the ideal embedded domain (Flash < 16 kB), while baselines need 13,185–157,500 params.

TABLE III: Cross-Dataset Comparative Evaluation

Model	Params	Dataset	MAE	RMSE
MicroPhys-BMS (Ours)	961	Stanford	0.536%	1.492%
		CALCE A123	0.173%	2.451%
PINN [27]	13,185	Stanford	0.898%	2.050%
		CALCE A123	0.137%	2.477%
Transformer [26]	13,899	Stanford	1.188%	2.367%
		CALCE A123	0.132%	2.484%
LightGBM Ens. [31]	$\approx 157,500$	Stanford	0.394%	1.037%
		CALCE A123	0.231%	1.299%

matching neural baselines with $13.7\times$ – $164\times$ fewer parameters.

2) Noise Robustness and CDF Analysis: As depicted in Fig. 6(a), MicroPhys-BMS achieves 93.83% CDF within $|\Delta\text{SOC}| \leq 2.0\%$ on Stanford, outperforming PINN (85.61%) and Transformer (79.33%). On CALCE A123, all models achieve > 97.5%. Under ± 5.0 mV noise, the proposed model maintains MAE = 1.343% (Stanford) and 0.249% (CALCE), both within the 1.456% target. As shown in Fig. 6(b), the 961-parameter Micro-MLP is the only model satisfying embedded constraints (Flash < 16 kB, ASIL-D) while maintaining competitive accuracy, whereas LightGBM requires 157,500 parameters.

IV. Conclusion

This study introduced MicroPhys-BMS, an end-to-end PIML-KD framework for real-time SOC estimation of LFP batteries on automotive microcontrollers. The core conclusions are:

- 1) Flat OCV Plateau Observability: The 12-D physical feature space X_{PIML} (ΔV_{10} , $\tau_{\log}$, P_{ohm} , Φ_{Arr} with $E_a = 35.0$ kJ/mol) resolves flat-plateau unobservability. The student Micro-MLP achieved MAE = 0.454% and RMSE = 1.236% ($R^2 = 0.9986$) at 25°C, outperforming its 300-tree LightGBM teacher (MAE = 0.488%).
- 2) Cross-Dataset Generalization: On Stanford LFP, the student recorded MAEs of 2.48% (YX07) and 3.35% (YX08), surpassing the Stanford baseline (4.30%)

[12] by 42.3% and 22.1%. On the independent CALCE A123 LFP cell [33] (different manufacturer, 18650 form), the student achieved MAE = 0.173% with 98.89% CDF compliance within $\leq 2.0\%$.

- 3) Comparative Evaluation: Against PINN [27] (13,185 params), Tabular Transformer [26] (13,899 params), and 5-model LightGBM ensemble [31] ($\approx 157,500$ params), the 961-parameter Micro-MLP achieved competitive accuracy (Stanford MAE 0.536% vs. 0.394%–1.188%; CALCE MAE 0.173% vs. 0.132%–0.231%) with $13.7\times$ – $164\times$ fewer parameters. It is the only model satisfying automotive embedded constraints (Flash < 16 kB, ASIL-D).
- 4) ASIL-D Deployment: The student compiled into `bms_soc_piml_mlp3.h` requires 3.75 kB Flash, 192 Bytes SRAM, zero malloc, and 12.0 μs per inference on ARM Cortex-M4 at 160 MHz, achieving $154.2\times$ latency acceleration over reference RF ensembles [23].

The complete source code, trained weights, and reproducible validation pipeline are available at <https://github.com/ha22da/PIML>. Future work will extend to online joint SOH/SOP co-estimation and integrate the deterministic C engine into commercial VCU energy management systems.

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